DIAGNOSTIC FLOWCHART · IDLE AND P0505

P0505 decision tree

P0505 stored? Save freeze frame before anyone clears it.

▼

Check idle RPM — warm, Park, loads off
Screen 700–800 rpm. TSB reference about 744 rpm. TEST REQUIRED

▼

High or unstable?

YES
Check vacuum leaks. Guide 04. Smoke and visual before any parts. A leak is a likely idle cause. It does not prove a lean mixture.

▼

Inspect the IAC
Both coils. Screen 40–80 Ω. Connector latched. O-ring present.

▼

Was the pintle moved by hand?
YES on this truck — stated. Treat the motor as damaged even if ohms are in range.

▼

Replace with a NEW correct IAC
Do not touch the new pintle. No used motor. Verify 4874373AB by VIN.

▼

Clean the throttle body
Plate must close. Cable must return.

▼

Reset PCM / idle relearn
After the leak is fixed. Guide 03. Foot off the pedal.

▼

Verify warm idle
700–800 rpm screen. Rescan. If still high, go back to smoke. Do not buy a second motor yet.

NO — idle is already in the screen
P0505 is then the electrical monitor until proven otherwise. A normal idle does not erase a shorted driver wire.

Electrical branch

Inspect the connector and wiggle the harness with the code being watched. Check both IAC coils. A short to ground or to battery on a driver wire is the Chrysler-family description of this code. LIKELY TEST REQUIRED

Compare all four circuits. Condemn the PCM only after the motor and the wires test good. Technician reports say a poor aftermarket IAC will bring this code back. POSSIBLE

Also look for P1294, target idle not reached, if the tool lists it. High idle with that code still sends you up the YES column.

Stop the tree

If the manifold glows at any box, key off. Leave this page. Guide 05. If oil is on the exhaust, Guide 06 before the road test that would “verify” the idle.

Unplugging the IAC freezes the pintle where it sits. It is a comparison, not a repair. A motor that was pushed does not get plugged back in for a second chance.

DIAGNOSTIC FLOWCHART · P0161 HEATER CIRCUIT

P0161 decision tree

Bank 2 Sensor 2. Rear downstream sensor, after the rear mini-cat, before the main cat. **LIKELY** Confirm the bung. This tree does not diagnose idle and it does not diagnose a glow.

P0161 stored?

▼

Inspect connector and wiring
Oil, corrosion, spread pins, melted insulation, harness touching the pipe. Cold engine.

▼

Check the fuse
ASD fuse from the under-hood lid. Measure it. Do not guess the cavity.

▼

Check heater resistance — sensor cold
Screen about 4–10 Ω on the sensor side. OL = open heater. Near 0 Ω = shorted heater.

▼

Check 12 V supply
About 10–12 V while the ASD is actually on. A steady key-on 0 V can be the relay already dropped out.

▼

Check PCM ground / control
The other heater wire. Often pulsed, not a hard ground. Compare with another sensor.

▼

Repair the wiring, fuse, or connector
OR replace the sensor if the heater element is open or shorted.

Oil gate

If oil is in this connector or on this sensor, fix the leak and dry the plug before a new sensor goes in. Otherwise you will buy this part twice. Guide 06.

Ohms	Power	Control	Do this
Open or short	Harness has feed	—	Sensor. After the oil gate.
4–10 Ω	Missing here only	—	Repair this feed wire.
4–10 Ω	Missing everywhere	—	Fuse or ASD relay.
4–10 Ω	10–12 V	Dead, others work	Control wire, then PCM.
4–10 Ω	10–12 V	Pulls / pulses	Intermittent connector or wire. Do not replace yet.

Not this code

A heater that tests good will not fix a high idle, a vacuum leak, or a glowing manifold. Signal codes are a different test. Do not swap sensors between banks. The pigtail lengths are not the same.

Parts gate NOT CONFIRMED

Do not order NTK 23142 or Mopar 56029001AB from this sheet. Neither checked out as the 2004 Grand Cherokee 4.0L Bank 2 Sensor 2. Guide 09. VIN and position at the parts counter.

Do not hard-ground the control wire and leave it. A heater around 5 Ω draws about 2.5 A. The PCM pulses it for a reason. No jumper repairs.